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### Gas spring-powered fastener driver with pressure mechanism

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#### Abstract

A gas spring-powered fastener driver includes a cylinder, a storage chamber cylinder having pressurized air in communication with the cylinder, a moveable piston positioned within the cylinder, a driver blade extending from a first side of the piston and movable therewith between a top-dead-center position and a bottom-dead-center position, a one-way seal carried onboard the piston and disposed between the piston and the cylinder, the one-way seal being configured permit a bypass flow of pressurized air from the first side of the piston, past the one-way seal, and into a space within the cylinder adjacent an opposite, second side of the piston during movement of the piston and driver blade from the top-dead-center position to the bottom-dead-center position, thereby increasing a pressure of the pressurized air within the cylinder and storage chamber cylinder, and a pressure relief valve in fluid communication with the storage chamber cylinder.

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**Inventors:** Bierdeman; David A. (New Berlin, WI), Thorson; Troy C. (Cedarburg, WI), Zimmerman; Jacob N. (Pewaukee, WI), Wyler; Andrew R. (Pewaukee, WI), Leathrum; Travis W. (Milwaukee, WI), Graf; David C. (Greendale, WI), Brasel; Alex J. (Greenfield, WI)

**Applicant:** MILWAUKEE ELECTRIC TOOL CORPORATION (Brookfield, WI)

**Family ID:** 84919661

**Assignee:** MILWAUKEE ELECTRIC TOOL CORPORATION (Brookfield, WI)

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*Primary Examiner:* Long; Robert F

*Attorney, Agent or Firm:* Michael Best & Friedrich LLP

## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of International Application No. PCT/US2022/037337, filed on Jul. 15, 2022, which claims priority to U.S. Provisional Patent Application No. 63/332,480, filed on Apr. 19, 2022, U.S. Provisional Patent Application No. 63/237,494, filed on Aug. 26, 2021, and U.S. Provisional Patent

## FIELD OF THE INVENTION

(1) The present invention relates to powered fastener drivers, and more specifically to gas spring-powered fastener drivers.

## BACKGROUND OF THE INVENTION

(2) There are various fastener drivers known in the art for driving fasteners (e.g., nails, tacks, staples, etc.) into a workpiece. These fastener drivers operate utilizing various means known in the art (e.g., compressed air generated by an air compressor, electrical energy, a flywheel mechanism, etc.), but often these designs are met with power, size, and cost constraints. One factor that existing fastener drivers do not account for relates to solving pressure loss over tool life and/or fluctuating pressure based on external temperatures. While an onboard air compressor may help alleviate pressure loss, typical air compressors (e.g., reciprocating, axial, screw, or centrifugal compressors) would add significant complexity, cost, and weight to the tool and therefore are unreasonable options.

## SUMMARY OF THE INVENTION

(3) The present invention provides, in one aspect, a gas spring-powered fastener driver including a cylinder, a storage chamber cylinder having pressurized air in communication with the cylinder, a moveable piston positioned within the cylinder, a driver blade extending from a first side of the piston and movable therewith between a top-dead-center position and a bottom-dead-center position, the driver blade defining a driving axis, a lifter operable to move the driver blade from the bottom-dead-center position toward the top-dead-center position, the lifter is configured to engage the driver blade when moving the driver blade from the bottom-dead-center position toward the top-dead-center position, a one-way seal carried onboard the piston and disposed between the piston and the cylinder, and a pressure relief valve in fluid communication with the storage chamber cylinder. The one-way seal being configured to permit a bypass flow of pressurized air from the first side of the piston, past the one-way seal, and into a space within the cylinder adjacent an opposite, second side of the piston during movement of the piston and driver blade from the top-dead-center position to the bottom-dead-center position, thereby increasing a pressure of the pressurized air within the cylinder and storage chamber cylinder. The pressure relief valve being in fluid communication with the storage chamber cylinder and configured to open in response to the pressure of the pressurized air within the storage chamber cylinder exceeding a predetermined value.

(4) The present invention provides, in another aspect, a gas spring-powered fastener driver including a cylinder, a storage chamber cylinder having pressurized air in communication with the cylinder, a moveable piston positioned within the cylinder, a driver blade attached to the piston and movable therewith between a top-dead-center position and a bottom-dead-center position, the driver blade defining a driving axis, a lifter operable to move the driver blade from the bottom-dead-center position toward the top-dead-center position, the lifter is configured to engage the driver blade when moving the driver blade from the bottom-dead-center position toward the top-dead-center position, a bumper positioned beneath the piston in a vertical direction to absorb impact energy from the piston; and a valve positioned in the storage chamber cylinder. The valve opens when a pressure of the pressurized air within the storage chamber cylinder exceeds a predetermined value.

(5) The present invention provides, in another aspect, a gas spring-powered fastener driver including a cylinder, a storage chamber cylinder having pressurized air in communication with the cylinder, a moveable piston positioned within the cylinder, a driver blade attached to the piston and movable therewith between a top-dead-center position and a bottom-dead-center position, the driver blade defining a driving axis, a lifter operable to move the driver blade from the bottom-

dead-center position toward the top-dead-center position, the lifter is configured to engage the driver blade when moving the driver blade from the bottom-dead-center position toward the top-dead-center position, and a sliding seal disposed between the piston and the cylinder. The sliding seal is configured to prevent pressurized air from passing through an annular space between the piston and the cylinder in only a single direction of movement of the piston along the driving axis. (6) Other features and aspects of the invention will become apparent by consideration of the following detailed description and accompanying drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is perspective view of a gas spring-powered fastener driver in accordance with embodiments of the invention.
- (2) FIG. 2 is a partial cross-sectional view of the gas spring-powered fastener driver taken along line 2-2 in FIG. 1.
- (3) FIG. 3 is a cross-sectional view of the gas spring-powered fastener driver of FIG. 1 taken along line 3-3 in FIG. 1, illustrating a motor and a transmission for providing torque to a lifter.
- (4) FIG. 4 is an enlarged cross-sectional view of a portion of the fastener driver illustrating a passageway for supplementing pressure in the fastener driver.
- (5) FIG. 5 is an enlarged cross-sectional view of a portion of the fastener driver similar to FIG. 4 and illustrating a check valve positioned in the passageway.
- (6) FIG. 6A is a schematic view of the gas spring-powered fastener driver of FIG. 1, illustrating a driver blade in a driven or bottom-dead-center position.
- (7) FIG. 6B is a schematic view of the gas spring-powered fastener driver of FIG. 1, illustrating a driver blade in a top-dead-center position prior to actuation.
- (8) FIG. 7 is a schematic of exemplary pressure relief in the fastener driver.
- (9) FIG. 8 is an enlarged cross-sectional view of a portion of the fastener driver illustrating a combined pressure-increase and pressure-relief mechanism in a first state.
- (10) FIG. 9 is another enlarged cross-sectional view of a portion of the fastener driver illustrating a combined pressure-increase and pressure-relief mechanism in a second state.
- (11) FIG. 10 is a cross-sectional view of the combined pressure-increase and pressure-relief mechanism in the second state.
- (12) FIG. 11 is a cross-sectional view of the combined pressure-increase and pressure-relief mechanism in the first state.
- (13) FIG. 12 is a schematic of a portion of a fastener driver including an exemplary glow plug that can be positioned in the outer storage chamber cylinder and that is in an operational state to account for a low operating temperature.
- (14) FIG. 13 is a schematic of the portion of the fastener driver of FIG. 12 including the glow plug in a non-operational state due to normal or high operating temperatures.
- (15) FIG. 14 is a cross-sectional view of the glow plug of FIG. 12.
- (16) FIG. 15 is a schematic of a portion of a fastener driver including an exemplary tank separator, illustrating a driver blade in a top-dead-center position with the tank separator in a neutral position.
- (17) FIG. 16 is a schematic of the tank separator of FIG. 15, illustrating the driver blade in a bottom-dead-center position with the tank separator in the neutral position at a first operating temperature.
- (18) FIG. 17 is a schematic of the tank separator of FIG. 15, illustrating the driver blade in a bottom-dead-center position with the tank separator in the neutral position at a second operating temperature.
- (19) FIG. 18 is a schematic of the tank separator of FIG. 15, illustrating the driver blade in the top-

dead-center position with the tank separator in a compressed position.

(20) FIG. **19A** is a schematic of another exemplary tank separator for a fastener driver.

(21) FIG. **19B** is a schematic of another exemplary tank separator for a fastener driver.

(22) FIG. **20** is a schematic of an exemplary bladder coupled to the outer storage chamber of a fastener driver, illustrating the bladder in a neutral position.

(23) FIG. **21** is a schematic of the bladder of FIG. **20**, illustrating the bladder in an expanded position.

(24) FIG. **22** is a schematic of an exemplary auxiliary tank coupled to the outer storage chamber of a fastener driver, illustrating a valve of the auxiliary tank in a closed position.

(25) FIG. **23** is a schematic of the auxiliary tank of FIG. **22**, illustrating the valve of the auxiliary tank in an open position.

(26) FIG. **24** is a schematic of an exemplary manual pressure adjuster coupled to the outer storage chamber of a fastener driver.

(27) FIG. **25** is a schematic of another exemplary manual pressure adjuster coupled to the outer storage chamber of a fastener driver, illustrating the manual pressure adjuster in a first or pressurized position.

(28) FIG. **26** is a schematic of the manual pressure adjuster of FIG. **25**, illustrating the manual pressure adjuster in a neutral position.

(29) FIG. **27** is a schematic of the manual pressure adjuster of FIG. **25**, illustrating the manual pressure adjuster in a second or depressurized position.

(30) FIG. **28** is a schematic of a portion of a fastener driver including an exemplary pump in the fastener driver, illustrating the pump in an extended or primed position.

(31) FIG. **29** is a schematic of the exemplary pump of FIG. **28**, illustrating the pump in a retracted or pump position.

(32) FIG. **30** is a schematic of a portion of a fastener driver including an exemplary driver blade that can be retracted different distances into the fastener driver.

(33) FIG. **31** is a schematic of a portion of a fastener driver including an exemplary pressure-controlled valve.

(34) FIG. **32** is a schematic of a portion of a fastener driver including an exemplary sealing member.

(35) FIG. **33** is a schematic of the sealing member of FIG. **32**.

(36) FIG. **34** is a schematic of a sealing member of a fastener driver.

(37) FIG. **35** is a perspective view of another exemplary cylinder.

(38) FIG. **36** is a cross-sectional view of the cylinder of FIG. **35**, showing a sealing member.

(39) FIG. **37** is a perspective view of another exemplary cylinder.

(40) FIG. **38** is a cross-sectional view of the cylinder of FIG. **37**, showing a sealing member.

(41) FIG. **39** is a perspective view of another exemplary cylinder.

(42) FIG. **40** is a cross-sectional view of the cylinder of FIG. **39**, showing a sealing member.

(43) FIG. **41** is a perspective view of another exemplary cylinder.

(44) FIG. **42** is a cross-sectional view of the cylinder of FIG. **41**, showing a sealing member.

(45) FIG. **43** is a perspective view of a portion of another exemplary gas-spring powered fastener driver.

(46) FIG. **44** is a cross-sectional view of the portion of the fastener driver of FIG. **43**, showing a pressure increase mechanism.

(47) FIG. **45A** is a schematic of the portion of the fastener driver of FIG. **43**, when a piston of the pressure increase mechanism is in a top-dead-center position.

(48) FIG. **45B** is a schematic of the fastener driver of FIG. **43**, when the piston is being pushed from a top-dead-center position to a bottom-dead-center position, such that air is compressed in an annular space.

(49) FIG. **45C** is a schematic of the fastener driver of FIG. **43**, when the piston has compressed air

within an annular space, expelling the compressed air into a storage chamber cylinder.

(50) FIG. 45D is a schematic of the fastener driver of FIG. 43, when the piston is being moved from the bottom-dead-center position to the top-dead-center position by a spring.

(51) FIG. 46 is a perspective view of another exemplary cylinder.

(52) FIG. 47 is a cross-sectional view of the cylinder of FIG. 46, showing a drive piston and attached driver blade at a top-dead-center position.

(53) FIG. 48 is a cross-sectional view of the cylinder of FIG. 46, showing the drive piston at a near bottom-dead-center position and upon initial contact with a bumper.

(54) FIG. 49 is an enlarged cross-sectional view of the cylinder of FIG. 46, showing a cylinder cap.

(55) FIG. 50 is a partial cross-sectional view of the cylinder of FIG. 46 through section line 50-50, showing the drive piston in a bottom-dead-center position and a one-way piston seal in an open state.

(56) FIG. 51 is a cross-sectional view of the cylinder of FIG. 46 though section line 51-51, showing a pressure relief valve.

(57) Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

#### DETAILED DESCRIPTION

(58) With reference to FIGS. 1-4, a gas spring-powered fastener driver 10 is operable to drive fasteners (e.g., single-headed nails, double-headed or duplex nails, tacks, staples, etc.) held within a magazine 14 into a workpiece. The fastener driver 10 includes an inner cylinder 18 and a moveable piston 22 positioned within the cylinder 18 (FIG. 5). With reference to FIG. 5, the fastener driver 10 further includes a driver blade 26 that is attached to the piston 22 and moveable therewith. The fastener driver 10 does not require an external source of air pressure, and instead includes an outer storage chamber cylinder 30 of pressurized fluid (e.g., gas) in communication with the cylinder 18. In the illustrated embodiment, the cylinder 18 and the movable piston 22 are positioned within the storage chamber cylinder 30.

(59) With reference to FIGS. 2, 4, and 5, the cylinder 18 and the driver blade 26 define a driving axis 38. As shown in FIGS. 6A and 6B, during a driving cycle the driver blade 26 and the piston 22 are moveable between a top-dead-center position (FIG. 6B) and a driven or bottom-dead-center position (FIG. 6A). FIG. 2 and illustrate that the fastener driver 10 further includes a lifter assembly 42 that is powered by a motor 46 and that is operable to move the driver blade 26 from the driven position to the top-dead-center position. The driver 10 also includes a latch assembly 48 that selectively holds the driver blade 26 in the ready position.

(60) In operation, the lifter assembly 42 drives the piston 22 and the driver blade 26 toward the top-dead-center position by energizing the motor 46. As the piston 22 and the driver blade 26 are driven toward the top-dead-center position, the gas above the piston 22 and the gas within the storage chamber cylinder 30 is compressed. Prior to reaching the top-dead-center position, the motor 46 is deactivated and the piston 22 and the driver blade 26 are held in a ready position, which is located between the top-dead-center and the bottom-dead-center or driven positions, until being released by user activation of a trigger 49. When released, the compressed gas above the piston 22 and within the storage chamber cylinder 30 drives the piston 22 and the driver blade 26 to the driven position, thereby driving a fastener into the workpiece. The illustrated fastener driver 10 therefore operates on a gas spring principle utilizing the lifter assembly 42 and the piston 22 to further compress the gas within the cylinder 18 and the storage chamber cylinder 30.

(61) With reference to FIGS. 2 and 6A-6B, the storage chamber cylinder 30 is concentric with the cylinder 18. The cylinder 18 has an annular inner wall 50 that guides the piston 22 and the driver



blade **26** along the driving axis **38** to compress the gas in the storage chamber cylinder **30**. The storage chamber cylinder **30** has an annular outer wall **54** circumferentially surrounding the inner wall **50**. The cylinder **18** has a threaded section **58** (FIG. 2). The storage chamber cylinder **30** has corresponding threads at a lower end **60** of the storage chamber cylinder **30** such that the cylinder **18** is threaded to the storage chamber cylinder **30** at the lower end **60**. As such, the cylinder **18** is configured to be axially secured to the storage chamber cylinder **30**.

(62) With reference to FIGS. 1 and 3, the driver **10** includes a housing **64** that has a cylinder support portion **68** in which the storage chamber cylinder **30** is at least partially positioned, and a motor support portion **72** in which the motor **46** and a transmission **80** are at least partially positioned. In the illustrated embodiment, the cylinder support portion **68** is integrally formed with the motor support portion **72** as a single piece (e.g., using a casting or molding process, depending on the material used). The transmission **80** raises the driver blade **26** from the driven position to the ready position. With reference to FIG. 3, the motor **46** is positioned within the transmission housing portion **72** to provide torque to the transmission **80** when activated. A battery pack **76** (FIG. 1) is electrically connectable to the motor **46** for supplying electrical power to the motor **46**. In some embodiments, the driver may be powered from an alternative power source such as an AC voltage input (i.e., from a wall outlet), or by an alternative DC voltage input (e.g., an AC/DC converter).

(63) With reference to FIGS. 3, the transmission **80** includes an input **84** (i.e., a motor output shaft) and includes an output shaft **96** that extends to a lifter **92** of the lifter assembly **42**, which is operable to move the driver blade **26** from the driven position to the ready position, as explained in greater detail below. In other words, the transmission **80** provides torque to the lifter **92** from the motor **46**. The illustrated transmission **80** is coupled to a transmission housing **100**. The transmission assembly **80** can take various forms and will not be described in detail. For example, the transmission assembly **80** may be the same as or similar to what is described in U.S. application Ser. No. 16/706,365 (titled "Gas Spring-Powered Fastener Driver"), the contents of which are incorporated by reference in their entirety. With reference to FIG. 2, the driver **10** further includes a lifter housing portion **106** (FIG. 2) positioned adjacent the storage chamber cylinder **30**. The lifter housing portion **106** substantially encloses the lifter assembly **42** that lifts the driver blade **26** to the ready position.

(64) With reference to FIG. 2, the fastener driver **10** includes a bumper **112** that is positioned beneath the piston **22** to stop the piston **22** at the driven position (FIG. 6A) and to absorb the impact energy from the piston **22**. The bumper **112** is configured to distribute the impact force of the piston **22** uniformly throughout the bumper **112** as the piston **22** is rapidly decelerated upon reaching the driven position (i.e., the bottom dead center position). The bumper **112** is disposed in the cylinder **18** and is clamped into place by the lifter housing portion **106**, which is threaded to the bottom end of the cylinder **18**. As shown, the bumper **112** is received within a cutout **114** that is formed in the lifter housing portion **106**. The cutout **114** coaxially aligns the bumper **112** relative to the driver blade **26**.

(65) In one example, and with reference to FIGS. 4 and 5, the fastener driver **10** includes a check valve **116** (or similar valve) that is positioned between the bumper **112** and the outer storage chamber cylinder **30** within a passageway **118**. The check valve **116** is responsive to pressure as the piston **22** compresses the bumper **112**. More specifically, as the piston **22** is driven from the ready position to the driven position, the piston **22** impacts the bumper **112**, which seals against the inner cylinder **18** to create an air reservoir or annular intermediate chamber **120**. The intermediate chamber **120** is formed between a bottom portion of the cylinder **18** and the bumper **112** (and in some circumstances, the bumper **112** and the piston **22**) when the driver blade **26** approaches the bottom-dead-center position. That is, the intermediate chamber **120** is completely sealed (i.e., not fluidly connected to the outside atmosphere) when the piston **22** impacts the bumper **112**. As the piston **22** compresses the bumper **112**, the pressure in the intermediate chamber **120** increases and

opens the check valve **116**. This increased air pressure through the opened check valve **116** adds a small amount of pressurized air to the outer storage chamber cylinder **30**, which results in a higher pressure applied to the cylinder **18** that can compensate for potential or actual air pressure losses in the driver **10**. As such, an increase in air pressure can be generated using bumper compression that occurs at the end of every firing event of the driver **10**. This avoids the need for a separate compressor to be attached to the cylinder **30** for increasing the pressure on the piston **22**. In effect, the complementary compression of the bumper **112** and the opening of the check valve **116** forms an onboard air compressor for the fastener driver **10**.

(66) By using the repetitive compression of the bumper **112** by the piston **22** to complement the pressure in the storage chamber cylinder **30**, a small amount of air pressure (e.g., approximately 0.01-0.015 psi) can be added each time the bumper **112** is compressed by the piston **22**.

Extrapolating this over 1000 nails fired by the driver **10**, this added pressure equates to approximately 10-15 psi, which is 10-15% of the total tank pressure. While the added pressure is relatively small compared to the total tank pressure, the added pressure facilitated by compression of the bumper **112** and the opened check valve **116** is enough to maintain an adequate tank pressure even after pressure losses are accounted for (e.g., due to permeation, minor debris ingress, or mild mechanical wear).

(67) In some circumstances, operational temperature associated with the fastener driver **10** or ambient temperature, or both, may increase the pressure applied to the piston **22** to an extent that a pressure relief is desirable. In these circumstances, and with reference to FIG. 7, the fastener driver **10** can include a pressure relief valve **124** that opens at a predetermined pressure to vent air when the pressure in the storage chamber cylinder **30** is higher than the pressure needed to correctly seat the fastener while also avoiding having the fastener driver **10** absorb more energy from movement of the piston **22** than is necessary. For example, at high temperatures, the pressure on the piston **22** may increase to an extent where air is vented via the valve **124** to keep the fastener driver **10** within a desired pressure tolerance range. In addition, in low operating temperatures for the fastener driver **10**, the onboard compressor defined by the compression of the bumper **112** and opening of the check valve **116** (i.e. leveraging the air reservoir formed by the bumper **112** when the bumper **112** seals against the cylinder **18**) assists with repressurizing the cylinder **18** to maintain performance of the fastener driver **10**.

(68) It will be appreciated that some embodiments of the fastener driver **10** may include, in combination, the check valve **116** to increase pressure within the storage chamber cylinder **30** and a pressure relief valve **124** that relieves pressure from the storage chamber cylinder **30**.

(69) In another example, and with reference to FIGS. 8-11, the fastener driver **10** may include a combination valve **128** that is positioned between the bumper **112** and the storage chamber cylinder **30** e.g., in the passageway **118** and that combines the functionality of a check valve and a pressure relief valve. An exemplary combination valve **128** is manufactured by Minivalve, Inc., located at 692 Oak Tree Boulevard, Suite 200, Cleveland, OH 44131. In this example, the valve **128** is responsive to pressure as the piston **22** compresses the bumper **112**, as well as responsive to the pressure in the storage chamber cylinder **30** such that the valve **128** can increase pressure in the storage chamber cylinder **30** when the pressure is lower than a desired amount. The valve **128** also can decrease or relieve pressure when the pressure in the storage chamber cylinder **30** is above a desired pressure for the piston **22**. As shown in FIGS. 10 and 11, the valve **128** includes a pressure release valve **132** and a membrane **136** that has a hole or opening **140** into which the pressure release valve **132** is coupled to or positioned within. The pressure release valve **132** includes a body that defines a tapered passageway **144** and that has an annular shoulder **148** and an annular flange **152**. The tapered passageway **144** extends narrower from a side of the pressure release valve **132** that is in communication with the air reservoir **120** toward the storage chamber cylinder **30**. The shoulder **148** is engaged with the membrane **136** on a first side within the storage chamber **30** to hold the pressure release valve **132** in place. The flange **152** is engaged with the membrane **136**

on a second side opposite the first side and is responsive to pressure within the storage chamber cylinder **30** to relieve pressure in the storage chamber cylinder **30** when the pressure is above a predetermined amount.

(70) The membrane **136** also includes apertures or openings **156** that facilitate pressure relief. The openings **156** are aligned with the flange **152** such that, as shown in FIG. **10**, when pressure in the storage chamber cylinder **30** is above a predetermined threshold, the excess pressure flips or inverts the flange **152** to relieve the excess pressure. After the pressure has been relieved, the flange **152** can be designed to return to the state shown. The pressure release valve **132** and the membrane **136** also facilitate increasing pressure within the storage chamber cylinder **30**. In particular, and with reference to FIG. **11**, when the piston **22** impacts the bumper **112** and generates additional air pressure within the air reservoir **120**, the additional high-pressure air is directed through the tapered passageway **144** into the storage chamber cylinder **30** to supplement the pressure in the storage chamber cylinder **30**. The tapered passageway **144** assists with providing airflow communication from the air reservoir **120** to the storage chamber cylinder **30** when it is desired to supplement the pressure in the storage chamber cylinder **30**, while also ensuring that pressure relief occurs via the openings **156** when the pressure in the storage chamber cylinder **30** is higher than desired (e.g., due to operational temperature, ambient temperature, or both).

(71) In another example, and with reference to FIGS. **12-14**, the fastener driver **10** includes a glow plug **200** positioned within the storage chamber cylinder **30**. As shown in FIGS. **12** and **14**, the glow plug **200** includes a sensor **202** that measures the pressure within the storage chamber cylinder **30**, and a heating rod **204** that selectively heats air within the storage chamber cylinder **30**. The sensor **202** may be a piezoresistive strain gauge pressure sensor or another sensor that can measure pressure. The glow plug **200** heats the air within the storage chamber cylinder **30** via the heating rod **204** based on the pressure measured by the sensor **202**.

(72) In use, the glow plug **200** can be used to heat the air within the storage chamber cylinder **30** when the pressure within the storage chamber cylinder **30** falls below a predetermined level. For example, the pressure within storage chamber cylinder **30** may drop as a result of a lower ambient temperature in the external environment. When the sensor **202** detects a decrease in pressure below the predetermined level, the glow plug **200** can be activated to heat the air within the storage chamber cylinder **30** and increase the pressure within the storage chamber cylinder **30** (FIG. **12**). With reference to FIG. **13**, when the sensor **202** determines that the pressure in the storage chamber cylinder **30** is above the predetermined level (e.g., due to an increase in temperature due to firing the fastener driver **10** or an elevated external ambient temperature), the glow plug is off (or turned off). In other words, the glow plug **200** is solely used to heat the air in the storage chamber cylinder **30** when the pressure within the storage chamber cylinder **30** is low. The storage chamber cylinder **30** may have a lower initial fill pressure than what would normally be used because the glow plug **200** can be implemented to increase the pressure. For example, the storage chamber cylinder **30** may be filled to approximately 80% of a typical initial fill pressure. The glow plug **200** regulates the pressure within the storage chamber cylinder **30** such that the pressure stays within a desired range regardless of the fire rate of the fastener driver **10** or the external ambient temperature.

(73) In another example, and with reference to FIGS. **15-18**, the fastener driver **10** includes a tank separator **300** disposed in the storage chamber cylinder **30** between an outer surface **304** of the cylinder **18** and an inner surface **306** of the storage chamber cylinder **30**. The illustrated tank separator **300** includes an annular ring **308** that spans the space between the cylinder **18** and the storage chamber cylinder **30**, and an O-ring that is coupled to the annular ring **308** at the outer surface **304** of the chamber **30**. In some embodiments, the annular ring **308** may be formed from a metal. In other embodiments, the annular ring **308** may be formed from other material (e.g., plastic, composite, etc.). The tank separator **300** is biased by a spring **312** toward the top-dead-center position (i.e., a neutral position of the tank separator **300**).

(74) As the temperature within the cylinder **18** and the storage chamber cylinder **30** increases, the

pressure within the cylinder **18** and the storage chamber cylinder **30** also increases. For example, and with reference to FIGS. **15-17**, the temperature within the cylinder **18** and the storage chamber cylinder **30** may increase as a result of firing the piston **22**. As the piston **22** moves downward, the tank separator **300** remains in the neutral position. FIG. **18** illustrates that, when the pressure within the cylinder **18** and the storage chamber cylinder **30** reaches a predetermined level due to added heat, the tank separator **300** is biased toward bottom-dead-center against the mechanical bias of the spring **312**. This movement of the tank separator **300** increases the volume of the storage chamber cylinder **30** to regulate the pressure within the storage chamber cylinder **30**. In other words, while the temperature within the storage chamber cylinder **30** increases, the pressure within the storage chamber cylinder **30** remains substantially constant (e.g., 164 psi) due to the increased volume of the storage chamber cylinder **30**.

(75) In some embodiments, the tank separator **300** may take the form of a gas spring that is coupled to the storage chamber cylinder **30**. In one example, and with reference to FIGS. **19A** and **19B**, the spring **312** may be located in a compartment **318** that protrudes outwardly from the storage chamber cylinder **30**. In some embodiments, the spring **312** may be a mechanical spring (FIG. **19A**). In other embodiments, the spring **312** may be a gas spring (FIG. **19B**). The compartment **318** is fluidly coupled to the storage chamber cylinder **30** via an opening **322**. The spring **312** biases the tank separator **300** toward the storage chamber cylinder **30**. As the pressure within the storage chamber cylinder **30** increases, the pressure biases the tank separator **300** away from the storage chamber cylinder **30**, compressing the spring **312**. Movement of the tank separator **300** increases the volume of the storage chamber cylinder **30**, which regulates the pressure within the storage chamber cylinder **30** to maintain a substantially constant pressure.

(76) In another example, and with reference to FIGS. **20** and **21**, the fastener driver **10** includes a bladder **400** that is fluidly coupled to the storage chamber cylinder **30**. The bladder **400** may be a rubber bladder, or the bladder **400** may include metal springs that act on a pressurized bladder portion of the bladder **400**. The bladder **400** is coupled to the storage chamber cylinder **30** via a conduit **404**. In some embodiments, the bladder **400** may be coupled to the storage chamber cylinder **30** via other connections. The bladder **400** varies from a neutral position (FIG. **20**) to an expanded position (FIG. **21**) to maintain substantially constant pressure (e.g., 164 psi) in the storage chamber cylinder **30**. In the neutral position, the bladder **400** has a first volume, and in the expanded position, the bladder **400** has a second, larger volume. In use, as the temperature within the storage chamber cylinder **30** increases, the pressure within the storage chamber cylinder **30** also increases. The increased air enters the bladder **400** via the conduit **404**, expanding the bladder **400** to the expanded position (FIG. **21**). Expansion of the bladder **400** increases the volume of the storage chamber cylinder **30** to account for the increase in pressure. The bladder **400** allows movement of air to and from the storage chamber cylinder **30** so that the pressure within the storage chamber cylinder **30** remains substantially constant regardless of the temperature within the storage chamber cylinder **30**.

(77) In another example, and with reference to FIGS. **22** and **23**, the fastener driver **10** includes an auxiliary tank **500** that is fluidly coupled to the storage chamber cylinder **30** via a pressure relief valve **504**. The pressure relief valve **504** includes a valve housing **508**, a plunger **512** disposed in the valve housing **508**, and a spring **516** that biases the plunger **512** toward the auxiliary tank **500**. In some embodiments, pressure relief valve **504** is a one-way valve such that air may solely travel from the auxiliary tank **500** to the storage chamber cylinder **30**. In other embodiments, the pressure relief valve **504** may be a two-way valve. The storage chamber cylinder **30** and the auxiliary tank **500** are filled with pressurized air to respective predetermined pressures (e.g., 164 psi and 500 psi, respectively). The respective pressures in the storage chamber cylinder **30** and in the auxiliary tank **500** may vary. The auxiliary tank **500** is filled to a higher pressure than the storage chamber cylinder **30** to accommodate a potential drop in pressure within the storage chamber cylinder **30**. When the pressure drops in the storage chamber cylinder **30** (e.g., due to a leak or a drop in

operating temperature caused by low ambient temperature), the pressure relief valve **504** opens to allow air from the auxiliary tank **500** to repressurize the storage chamber cylinder **30**. More specifically, the spring **516** and the air within the storage chamber cylinder **30** bias the plunger **512** toward a closed position, against the bias of the air within the auxiliary tank **500**. When the pressure within the storage chamber cylinder **30** drops, so does the force acting on the plunger **512**. This drop in force allows the air within the auxiliary tank **504** to bias the plunger **512** to an open position, allowing the air from the auxiliary tank **504** to enter the storage chamber cylinder **30**, repressurizing the storage chamber cylinder **30**. The pressure relief valve **504** may repressurize the storage chamber cylinder **30** to a minimum pressure. In some embodiments, the minimum pressure may be approximately 140 psi. In other embodiments, the minimum pressure may be higher or lower than 140 psi.

(78) In another example, and with reference to FIG. **24**, the fastener driver **10** includes a manual pressure adjuster **600** located at an end of the storage chamber cylinder **30**. In other embodiments, the location of the manual pressure adjuster **600** may differ. The manual pressure adjuster **600** has an adjustable portion **604** that is coupled to the storage chamber cylinder **30** (e.g., by threaded connection). A user may engage the adjustable portion **604** to alter the position of the adjuster **600** relative to the storage chamber cylinder **30**. Altering the position of the adjustable portion **604** changes the volume of the storage chamber cylinder **30** so that the pressure within the storage chamber cylinder **30** can be set to or maintained at a desired level (e.g., 164 psi). For example, rotating or otherwise moving the adjustable portion **604** a small amount marginally changes the volume of (and therefore the pressure in) the storage chamber cylinder **30**, whereas rotating or moving the adjustable portion **604** a relatively large amount changes the volume of (and therefore the pressure in) the storage chamber cylinder **30** a correspondingly large amount. The adjuster **600** can be used to maintain the pressure substantially constant within the storage chamber cylinder **30**, or to restore pressure that has been lost. For example, the volume within the storage chamber cylinder **30** can be increased by rotating the adjustable portion **604** outward (upward in FIG. **24**) to account for higher temperatures that increase the pressure in the storage chamber cylinder **30**. The temperature affecting the pressure within the storage chamber cylinder **30** may be high as a result of firing the fastener driver **10** or a high ambient temperature.

(79) The adjuster **600** may take different forms. For example, and with reference to FIGS. **25-27**, the adjuster **600** may be movable between a plurality of predetermined positions. For example, the adjustable portion **604** may be movable between a neutral position (FIG. **26**), a first or pressurized position (FIG. **25**), and a second or depressurized position (FIG. **27**). In other embodiments, the adjustable portion **604** may be movable between more than three positions or fewer than three positions. Each of the first and second positions alters the volume of the storage chamber cylinder **30** relative to the neutral position. For example, the second position increases the volume of the storage chamber cylinder **30** (FIG. **27**), whereas the first position decreases the volume of the storage chamber cylinder **30** (FIG. **25**). The neutral position maintains the volume of the storage chamber cylinder **30** (FIG. **26**). The adjuster **600** may include one or more detents **608** that interact with corresponding protrusions **612** in the storage chamber cylinder **30** (or vice versa) to hold the adjustable portion **604** at each position, as shown in FIG. **24**. In other embodiments, the adjustable portion **604** may be held to the storage chamber cylinder **30** in other ways.

(80) In use, the user may move the adjustable portion **604** to vary the pressure within the storage chamber cylinder **30**. The storage chamber cylinder **30** is filled with compressed air to a predetermined pressure when the adjustable portion **604** is in the neutral position. When the pressure within the storage chamber cylinder **30** is low, the user can move the adjustable portion **604** to the first position (FIG. **25**), which decreases the volume of the storage chamber cylinder **30** and increases the pressure within the storage chamber cylinder **30**. When the pressure within the storage chamber cylinder **30** is high, the user can move the adjustable portion **604** to the second position (FIG. **27**), which increases the volume for the storage chamber cylinder **30** and decreases

the pressure within the storage chamber cylinder **30**.

(81) In some embodiments, the manual pressure adjuster **600** may include an indicator **616**, as shown in FIG. **1**. The indicator **616** can be positioned on an exterior surface of the fastener driver **10** such that the indicator **616** is visible to the user. The indicator **616** may indicate to the user that moving the adjustable portion **604** to the first position provides a deeper seating of the fastener in the workpiece and moving the adjustable portion **604** to the second position provides a shallower seating of the fastener in the workpiece. The seating of the fastener is influenced by the pressure of the air within the storage chamber cylinder **30**. The indicator **616** may additionally or alternatively indicate to the user that the pressure within the storage chamber cylinder **30** is within a predetermined range. In some embodiments, a separate indicator may indicate the pressure within the storage chamber cylinder **30**. In other embodiments, the indicator **616** may solely indicate the seating of the fastener. In other embodiments, the indicator **616** may solely indicate the pressure within the storage chamber cylinder **30**.

(82) In another example, and with reference to FIGS. **28** and **29**, the fastener driver **10** includes a pressure regulating system **700**. The pressure regulating system **700** includes first and second check valves **702**, **704** in a wall **708** of the cylinder **18** beneath the bumper **112**. When open, the first and second check valves **702**, **704** fluidly couple the cylinder **18** to the storage chamber cylinder **30**. The pressure regulating system **700** additionally includes a third check valve **712** that is disposed on a wall **716** of the storage chamber cylinder **30**. When open, the third check valve **712** fluidly couples the storage chamber cylinder **30** to external atmosphere.

(83) In use, the temperature within the cylinder **18** and the storage chamber cylinder **30** increases when the piston **22** impacts the bumper **122**. This increase in temperature increases the pressure within the cylinder **18**. When the pressure within the cylinder **18** reaches a predetermined level, the first and second check valves **702**, **704** open to allow pressurized air to enter the storage chamber cylinder **30**. The flow of pressurized air into the storage chamber cylinder **30** increases the pressure in the storage chamber cylinder **30**. When the pressure in the storage chamber cylinder **30** increases beyond a predetermined level, the third check valve **712** opens to the external environment to depressurize the storage chamber cylinder **30** at least partially. As such, the pressure within the storage chamber cylinder **30** is regulated so that the pressure does not exceed a predetermined level. Bleeding the air from the storage chamber cylinder **30** is advantageous in situations when the fastener driver **10** has a high fire rate, when the external ambient temperature is high, or when both factors are present.

(84) In another example, and with reference to FIG. **30**, the fastener driver **10** includes a driver blade **800** that can be retracted farther than existing driver blades to change the pressure in the storage chamber cylinder **30** that impacts the piston on subsequent strokes. The increased retraction may require an increase in tool height (e.g., approximately 1-2 inches). In this embodiment, the fastener driver **10** includes a lifter **804** that is smaller than a lifter in the lifter assembly **42**. The smaller lifter **804** rotates more than a larger lifter to allow incremental adjustment to the pressure within the storage chamber cylinder **30**. In some embodiments, the lifter **804** has a diameter of approximately 0.15 inches. In some embodiments, the lifter **804** turns between 16 and 18 times to move the piston **22** from the bottom-dead-center position to the top-dead-center position. By rotating the lifter **804** additional revolutions, the distance between top-dead-center and bottom-dead-center increases, which allows the piston **22** to compress the pressurized air within the storage chamber cylinder **30** to a higher pressure when firing. In this example, the storage chamber cylinder **30** initially may be filled to a lower pressure because the piston **22** compresses the air to a higher pressure. For example, in some embodiments the initial fill pressure of the storage chamber cylinder **30** can be 80% of the normal fill pressure.

(85) Additionally, the driver blade **800** includes multiple notches **806** (only one of which is shown in FIG. **30**) that interact with a pawl **808** to hold the driver blade **800** in different positions depending on how far the driver blade **800** is lifted by the lifter **804**. For example, the driver blade

**800** may be held in a low-power position **812** (a first top-dead-center position), a medium-power position **816** (a second top-dead-center position), and a high-power position **820** (a third top-dead-center position). In the low-power position **812** of the illustrated driver blade **800**, the lifter **804** rotates 16 times from the bottom-dead-center position. In the medium-power position **816**, the lifter **804** rotates 17 times from the bottom-dead-center position. In the high-power position **820**, the lifter **804** rotates 18 times from the bottom-dead-center position.

(86) In another example, and with reference to FIG. 31, the fastener driver **10** includes a valve **900** that is disposed in the storage chamber cylinder **30**. The illustrated valve **900** is positioned at an end **904** of the cylinder **18** such that a sleeve **908** extending from a body **912** of the valve **900** engages outer surfaces **916** of the cylinder **18**. The valve **900** includes a leg **920** that extends from the body **912** of the valve **900** through an end **924** of the storage chamber cylinder **30** such that the leg **920** is exposed to the external environment. A spring **928** is positioned around the leg **920** to bias the body **912** toward the cylinder **18**. In use, an increased pressure within the cylinder **18** pushes the valve **900** toward the end **924** of the storage chamber cylinder **30**, compressing the spring **928**.

Movement of the valve **900** in this direction (upward in FIG. 31) reduces the air pressure applied to the piston **22** from the storage chamber cylinder **30** by decreasing the flow rate of air toward the piston **22**. As the pressure increases or decreases, the sleeve **908** moves in response to adjust the flow rate of air acting on the piston **22**. After the pressure normalizes, the spring **928** biases the sleeve **908** to the nominal position toward the piston **22**. The preload on the spring **928** may be adjusted to change the power output of the driver blade.

(87) In another example, and with reference to FIGS. 32 and 33, the fastener driver **10** includes a sealing member **1010** that is disposed between the piston **22** and the cylinder **18**. The sealing member **1010** seals an annular space between the piston **22** and the cylinder **18** such that compressed air does not escape through the annular space. In other words, the sealing member **1010** prevents compressed air from escaping between the piston **22** and an inner wall **1014** of the cylinder **18** as the piston **22** moves between the top-dead-center position and the bottom-dead-center position.

(88) The check valve **116** may be included in the present embodiment, as shown in FIG. 5. The check valve **116** is responsive to pressure as the piston **22** moves from top-dead-center to bottom-dead-center. More specifically, as the piston **22** is driven from the ready position to the driven position, the sealing member **1010** prevents compressed air from escaping from the intermediate chamber **120**. As pressure within the intermediate chamber **120** builds, the check valve **116** is opened, allowing compressed air to flow from the intermediate chamber **120** into the outer storage chamber cylinder **30**, repressurizing the outer storage chamber cylinder **30**. In other embodiments, the fastener driver **10** may not include the check valve **116**.

(89) The sealing member **1010** includes a ring **1018** and a wall **1022** integrally formed with the ring **1018**. The ring **1018** includes a first face **1026** and a second face **1030** that is parallel with the first face **1026**. The ring **1018** additionally includes a circular cutout **1034** positioned at a center of the ring **1018**. The wall **1022** extends outward from an edge of the ring **1018** such that inner and outer faces **1038**, **1042** of the wall **1022** are at obtuse angles relative to the first and second faces **1026**, **1030** of the ring **1018**. For example, the outer faces **1038**, **1042** may be at a 95-degree angle, a 100-degree angle, or a similar angle relative to the first and second faces **1026**, **1030** of the ring **1018**. In other embodiments, the outer faces **1038**, **1042** may be at an angle equal to or less than 90 degrees relative to the first and second faces **1026**, **1030** of the ring **1018**. The wall **1022** is continuous along the edge of the ring **1018** such that the wall **1022** also forms an annular ring. An edge of the wall **1022** is chamfered such that the inner face **1038** has a length or height that is shorter than a length or height of the outer face **1042**. When positioned within the fastener driver **10**, the ring **1018** extends inward into a space formed in the piston **22**, and the wall **1022** is positioned along a surface **1046** of the piston **22**. The inner face **1038** of the wall **1022** is in contact with the surface **1046** of the piston **22** while the outer face **1042** of the wall **1022** is in contact with the inner wall

**1014** of the cylinder **18**. The sealing member **1010** may be formed of a rubber material, a silicone material, or the like. The sealing member **1010** acts as a single-acting seal. In other words, the sealing member **1010** holds pressure in only one direction.

(90) In another example, and with reference to FIG. **34**, the sealing member **1010** may include first and second arms **1110**, **1114** which extend from the edge of the wall **1022**. In other words, the first and second arms **1110**, **1114** replace the chamfer shown in FIG. **33**. The first arm **1110** includes a first face **1118** that is continuous with the inner face **1038**. The second arm **1114** includes a second face **1122** that is continuous with the outer face **1042**. The first and second arms **1110**, **1114** are angled away from each other such that the first face **1110** is angled relative to the inner face **1038** and the second face **1114** is angled relative to the outer face **1042**. The second arm **1114** defines a length that is greater than a length of the first arm **1110**. The second arm **1114** additionally defines a width that is greater than a width of the first arm **1110**. In other embodiments, the length and/or the width of the second arm **1114** may be less than or the same as the length and/or the width of the first arm **1110**. Cross sectional shapes of the arms may be rectangular, circular, oblong, or the like. When positioned within the fastener driver **10**, the first face **1118** of the first arm **1110** is in contact with the surface **1046** of the piston **22** while the second face **1122** of the second arm **1114** is in contact with the inner wall **1014** of the cylinder **18**.

(91) In another example, and with reference to FIGS. **35** and **36**, the fastener driver **10** includes a pressure release mechanism **1200** disposed between the cylinder **18** and the storage chamber cylinder **30**. The cylinder **18** includes an aperture **1204** extending between the cylinder **18** and the storage chamber cylinder **30** such that the cylinder **18** and the storage chamber cylinder **30** are fluidly connected. The pressure release mechanism **1200** is positioned proximate the aperture **1204** to seal the aperture **1204**. In other words, the pressure release mechanism **1200** seals the aperture **1204** such that the cylinder **18** and the storage chamber cylinder **30** are fluidly distinct. In some embodiments, the pressure release mechanism **1200** is a slidable plug **1208** with an O-ring **1212** disposed on an exterior of the slidable plug **1208**. A spring **1216** is disposed on an end of the slidable plug **1208** such that the slidable plug **1208** is biased toward the aperture **1204**. When the pressure within the cylinder **18** reaches a predetermined pressure, the slidable plug **1208** is pushed away from the cylinder **18**, compressing the spring **1216**. Movement of the slidable plug **1208** allows air to escape from the cylinder **18** and into the storage chamber cylinder **30**.

(92) In other embodiments, as shown in FIGS. **37** and **38**, the pressure release mechanism **1200** is an O-ring **1300**. The O-ring **1300** sits in an angled groove **1304** on the surface of the cylinder **18**, proximate the aperture **1204**. The O-ring **1300** seals an opening **1306** to the aperture **1204**, such that fluid cannot exit the opening **1306** of the aperture **1204** without movement of the O-ring **1300**. The O-ring **1300** is formed from an elastic material such that the O-ring **1300** flexes when stretched. When the pressure within the cylinder **18** reaches a predetermined level, the pressure within the cylinder **18** stretches the O-ring **1300** such that air may escape from the cylinder **18**, into the storage chamber cylinder **30**. In other embodiments, as shown in FIGS. **39** and **40**, the O-ring **1300** may be replaced with a band **1400**. Similar to the O-ring **1300**, the band **1400** seals the opening **1306** of the aperture **1204**. The band **1400** is formed from an elastomeric material such that the band **1400** is flexible. In some embodiments, a band clamp **1404** may be positioned on a portion of the band **1400** to maintain the position of the band **1400** on the surface of the cylinder **18**. In other embodiments, the band clamp **1404** may not be included.

(93) In other embodiments, as shown in FIGS. **41** and **42**, the pressure release mechanism **1200** is U-seal **1500** that sits proximate the opening **1306** of the aperture **1204** such that the opening **1306** is sealed. The U-seal **1500** is formed from an elastic material such that the U-seal **1500** flexes when biased. When the pressure within the cylinder **18** reaches a predetermined level, the pressure within the cylinder **18** flexes the U-seal **1500** such that a portion of the U-seal **1500** seals against the surface of the cylinder **18**, rather than the opening **1306** of the aperture **1204**. In this position, air can escape from the cylinder **18** to the storage chamber cylinder **30**. When the pressure within the



cylinder **18** is below a predetermined level, the U-seal **1500** again seals against the opening **1306** of the aperture **1204**.

(94) In another example, and with reference to FIGS. **43** and **44**, the lifter assembly **42** includes a cam **1600** that engages with a small piston **1604**. More specifically, when the piston **22** is at the bottom-dead-center position, the lifter assembly **42** urges the cam **1600** to engage with the small piston **1604**. The lifter assembly **42** disengages the cam **1600** and the small piston **1604** as the piston moves from the bottom-dead-center position to top-dead-center position. In some embodiments, the cam **1600** may instead engage with a feature on the piston **22**. The small piston **1604** is positioned and slidable within an aperture **1608** in the storage chamber cylinder **30**. The cam **1600** is configured to bias the small piston **1604** from a small piston top-dead-center position to a small piston bottom-dead-center position, as shown in FIG. **45A**. In other words, the cam **1600** biases the small piston **1604** toward the storage chamber cylinder **30**. As the cam **1600** biases the small piston **1604** from the small top-dead-center position to the small bottom-dead-center position, the small piston **1604** compresses air in the aperture **1608**, increasing the pressure within the aperture **1608**, as shown in FIG. **45B**.

(95) The aperture **1608** includes a check valve **1612** positioned proximate the small bottom-dead-center position. Once the pressure within the aperture **1608** reaches a predetermined pressure, the check valve **1612** opens, allowing air to enter the storage chamber cylinder **30**, increasing the pressure within the storage chamber cylinder **30**, as shown in FIG. **45C**. The lifter assembly **42** releases the small piston **1604** when the small piston **1604** is in the small bottom-dead-center position. Once the lifter assembly **42** releases the small piston **1604**, a spring **1616** on the small piston **1604** biases the small piston **1604** back to the small top-dead-center position, as shown in FIG. **45D**. As the small piston **1604** moves to the small top-dead-center position, a vacuum is created within the aperture **1608**. A hole **1620** in the aperture **1608** allows external air to enter the aperture **1608**. Thereafter, the small piston **1604** is again urged to the small bottom-dead-center position by the lifter assembly **42**, starting an additional compression cycle. The storage chamber cylinder **30** includes a bleed valve **1624** that allows air to bleed out of the storage chamber cylinder **30** and into the external environment when a predetermined pressure is reached. In other embodiments, the storage chamber cylinder **30** may not include a bleed valve.

(96) In another example, and with reference to FIGS. **46-49**, the fastener driver **10** includes a cap **1650** attached to a cylindrical mount **1654** of the cylinder **18**. The cap **1650** is configured to prevent “piston pumping” from occurring within the cylinder **18** during each stroke of the drive piston **22**. “Piston pumping” increases the amount of compressed air stored inside the storage chamber cylinder **30** of the fastener driver **10**, which increases the pressure of the compressed air inside the cylinder **30**. Specifically, piston pumping can occur in two stages during the stroke of the drive piston **22**. As shown in FIGS. **47** and **48**, a first compression stage can occur when the drive piston **22** moves from top dead center position to a near bottom dead center position in which the piston **22** initially contacts the bumper **112**, shown in FIG. **48**. During this time, air at atmospheric pressure beneath the drive piston **22** is exhausted from the inner cylinder **18** and through vents (not shown) within the cap **1650** leading to the exterior of the fastener driver **10** (and ambient atmosphere). If these vents are too small in cross-sectional area, a high-pressure drop is created at the vents, preventing the pressurized exhaust air from escaping through the vents. Instead, this exhaust air may be pumped past the drive piston **22** and into the space in the inner cylinder **18** above the drive piston **22**, increasing the amount and pressure of compressed air in the storage chamber cylinder **30**.

(97) As shown in FIG. **50**, a second compression stage of piston pumping can occur after the piston **22** contacts the bumper **112**. At this time, a temporary seal **1656** may occur as the bumper **112** deflects outward. The temporary seal **1656** further inhibits the exhaust airflow from escaping to atmosphere through the vents in the cap **1650**. Instead, the exhaust airflow becomes trapped between the drive piston and the temporary seal, and if the pressure of the trapped exhaust air

exceeds the pressure in the inner cylinder **18** above the drive piston **22**, the exhaust airflow may backflow into the storage chamber cylinder **30** (indicated by arrows **1711** in FIG. **50**) instead of escaping into the environment.

(98) In some embodiments, and with reference to FIG. **48**, the vents **1658** within the cap **1650** are evenly disposed around a circumference of the mount **1654**. In some embodiments, eight of the vents **1658** may be evenly spaced along a circumference of the mount **1654**, creating a sufficiently large total cross-sectional area through which the exhaust airflow may be discharged during the downward stroke of the drive piston **22** to reduce or prevent the first stage of piston pumping described above. In other embodiments, more than or less than eight of the vents **1658** may be used. The vents **1658** allow the exhaust airflow beneath the drive piston **22** to escape from the fastener driver **10**, preventing the first stage of piston pumping.

(99) In some embodiments, and with reference to FIG. **49**, the cap **1650** also includes a guide slot **1662** through which the driver blade **26** extends during the downward stroke of the drive piston **22**. If the cross-sectional area of the guide slot **1662** is formed too small to inhibit ingress of debris into the mount **1654** and inner cylinder **18**, the magnitude of piston pumping during the first stage may be increased. To counter this, additional vents **1666** are formed disposed on the cap between a cap ring **1670** and a cap body **1674** to reduce piston pumping during the first stage. In some embodiments, four vents **1666** may be formed in the cap **1650**. In other embodiments, more than or less than four vents **1666** may be formed in the cap **1650**.

(100) In some embodiments, and with reference to FIGS. **47**, **48**, and **50**, the fastener driver **10** includes a one-way seal **1710** that is disposed between the piston **22** and the cylinder **18**. The one-way seal **1710** seals an annular space between the piston **22** and the cylinder **18** such that compressed air does not escape through the annular space. In other words, the one-way seal **1710** prevents compressed air from escaping between the piston **22** and an inner wall **1712** of the cylinder **18** as the piston **22** moves between the top-dead-center position and the bottom-dead-center position.

(101) As shown in FIG. **50**, the one-way seal **1710** is configured to intentionally permit a bypass flow **1711** of pressurized air from a first side **1713** of the piston **22**, on which the driver blade **26** extends therefrom, through the annular space, and into a space within the cylinder adjacent an opposite, second side **1714** of the piston **22**. The bypass flow **1711** moves through the annular space during movement of the piston **22** and the driver blade **26** from the top-dead-center position to the bottom-dead-center position, during the second stage of piston pumping described above, such that a pressure of the pressurized air within the cylinder **18** and the storage chamber cylinder **30** is increased to replace lost or leaked air from the cylinder **30**. In other words, when a pressure on the first side **1713** of the piston **22** is sufficiently high and the pressure on the second side **1714** of the piston **22** is sufficiently low, the bypass flow **1711** is induced through the annular space and past the one-way seal **1710**. However, the one-way seal **1710** does not permit the bypass flow **1711** to move in an opposite direction. For example, the bypass flow **1711** may solely flow from the first side **1713** of the piston **22** to the second side **1714** of the piston **22**, and not from the second side **1714** of the piston **22** to the first side **1713** of the piston **22**. If the pressure of the pressurized air within the cylinder **18** and the storage chamber cylinder **30** is sufficiently high, a pressure differential between the first side **1713** of the piston **22** and the second side **1714** of the piston **22** will be low. In such a case, the bypass flow **1711** will not be induced, and the one-way seal **1710** will remain closed.

(102) The one-way seal **1710** includes a ring **1718** and a wall **1722** integrally formed with the ring **1718**. The wall **1722** is continuous along the edge of the ring **1718** such that the wall **1722** also forms an annular ring. When positioned within the fastener driver **10**, the ring **1718** extends inward into a space **1723** (e.g., a groove) formed in the piston **22**, and the wall **1722** is positioned along the inner wall **1712** of the cylinder **18**. The one-way seal **1710** may be formed of a rubber material, a silicone material, or the like. The one-way seal **1710** holds pressure in only one direction.

(103) With continued reference to FIG. 50, the one-way seal **1710** further includes first and second arms **1810**, **1814** that extend from the ring **1718**. The first arm **1810** includes a first face **1818** that is in abutting contact with the wall **1722** of the drive piston **22**. The second arm **1814** includes a second face **1822** that may be in sliding contact with the inner wall **1712**. The first and second arms **1810**, **1814** are angled away from each other such that the first face **1810** is angled relative to the wall **1722** and the second face **1814** is angled relative to the inner wall **1712**. The second arm **1814** defines a length that is the same as a length of the first arm **1810**. The first and second arms **1810**, **1814** additionally define thicknesses that are tapered. A U-shaped recess **1826** is defined between the first arm **1810** and the second arm **1814**. When positioned within the fastener driver **10**, the first face **1818** of the first arm **1810** is in contact with a surface of the piston **22** while the second face **1822** of the second arm **1814** is in contact with the inner wall **1712** of the cylinder **18**. When the bypass flow **1711** of the pressurized air is induced in the annular space, the pressurized air causes the second arm **1814** to move toward the first arm **1810**, thereby allowing the bypass flow **1711** to flow between the piston **22** and the cylinder **18**, as shown in FIG. 50. When the annular space is created, the pressurized air may move from the first side **1712** of the piston **22**, to the space on the second side **1714** of the piston **22**.

(104) With reference to FIG. 51, the fastener driver **10** includes a pressure relief valve **124** in fluid communication with the interior of the storage chamber cylinder **30**. The pressure relief valve **124** is configured to open in response to the pressure of the pressurized air within the storage chamber cylinder **30** exceeding a predetermined value. The pressure relief valve **124** includes a seat **1913** disposed in a passageway **1914** of the storage chamber cylinder **30**, a pin **1915** disposed in the passageway **1914**, and a spring **1916** positioned between the seat **1913** and the pin **1915** to bias the pin **1915** toward the storage chamber cylinder **30** to seal the passageway **1914**. The seat **1913** is coupled to the passageway **114** via threads in the passageway **1914**. The seat **1913** is visible externally from the fastener driver **10**. The pin **1915** is slidable in the passageway **1914** between a closed position and an open position. The pin **1915** is moveable from the closed position to the open position when the pressure of the pressurized air within the storage chamber cylinder **30** exceeds the predetermined value. In the closed position, the pressure of the pressurized air is at or below the predetermined value, allowing the spring **1916** to bias the pin **1915** away from the seat **1913**. In the open position, the pressure of the pressurized air within the storage chamber cylinder **30** is above the predetermined value, such that the pressurized air biases the pin **1915** toward the seat **1913**, against the bias of the spring **1916**. In some embodiments, a refill valve **1920** may be disposed proximate the pressure relief valve **124**. The refill valve **1920** allows a user to add pressurized air to the storage chamber cylinder **30**.

(105) The pressure relief valve **124** opens at a predetermined pressure value to vent air when the pressure in the storage chamber cylinder **30** is higher than the pressure needed to correctly seat the fastener, while also avoiding having the bumper **112** absorb more energy from movement of the piston **22** than is necessary. For example, at high temperatures, the pressure on the piston **22** may increase to an extent where air is vented via the valve **124** to keep the fastener driver **10** operating within a desired range of operating pressures. In addition, in low operating temperatures for the fastener driver **10**, piston pumping during the second stage assists with repressurizing the cylinder **30** to maintain the fastener driver **10** within a desired range of operating pressures.

(106) Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described.

## Claims

1. A gas spring-powered fastener driver comprising: a cylinder; a storage chamber cylinder having pressurized gas in communication with the cylinder; a moveable piston positioned within the

cylinder; a driver blade extending from a first side of the piston and movable therewith between a top-dead-center position and a bottom-dead-center position, the driver blade defining a driving axis; a lifter operable to move the driver blade from the bottom-dead-center position toward the top-dead-center position, the lifter is configured to engage the driver blade when moving the driver blade from the bottom-dead-center position toward the top-dead-center position; a one-way seal carried onboard the piston and disposed between the piston and the cylinder, the one-way seal being configured permit a bypass flow of pressurized gas from the first side of the piston, past the one-way seal, and into a space within the cylinder adjacent an opposite, second side of the piston during movement of the piston and driver blade from the top-dead-center position to the bottom-dead-center position, thereby increasing a pressure of the pressurized gas within the cylinder and storage chamber cylinder; and a pressure relief valve in fluid communication with the storage chamber cylinder and configured to open in response to the pressure of the pressurized gas within the storage chamber cylinder exceeding a predetermined value, wherein the one-way seal includes a ring and an arm in sliding contact with the cylinder, the arm being integrally formed with and extending from the ring, wherein the ring extends inward into a groove formed in the piston, and wherein the arm is movable relative to an outer circumferential surface of the piston.

2. The gas spring-powered fastener driver of claim 1, wherein the one-way seal is configured to prevent pressurized gas from passing through an annular space between the piston and the cylinder.

3. The gas spring-powered fastener driver of claim 2, wherein the one-way seal allows pressurized gas to pass through the annular space in a first direction and prevents pressurized gas from travelling through the annular space in an opposite, second direction.

4. The gas spring-powered fastener driver of claim 3, wherein the bypass flow passes through the annular space when the driver blade reaches the bottom-dead-center position.

5. The gas spring-powered fastener driver of claim 1, wherein the arm is a first arm, wherein the one-way seal includes a second arm, and wherein the first arm and the second arm are angled away from each other.

6. The gas spring-powered fastener driver of claim 5, wherein the bypass flow of pressurized gas causes the first arm to move toward the second arm, thereby creating an annular space between the piston and the cylinder.

7. The gas spring-powered fastener driver of claim 1, wherein the pressure relief valve includes: a seat positioned within a passageway in the storage chamber cylinder fluidly communicating an interior of the storage chamber cylinder with the ambient surroundings of the fastener driver, a pin disposed in the passageway, and a spring positioned between the seat and the pin to bias the pin toward the storage chamber cylinder to seal the passageway.

8. The gas spring-powered fastener driver of claim 7, wherein when the pressure of pressurized gas within the storage chamber cylinder exceeds the predetermined value, the pin is moved away from the storage chamber cylinder, against the bias of the spring, to open the passageway.

9. A gas spring-powered fastener driver comprising: a cylinder; a storage chamber cylinder having pressurized gas in communication with the cylinder; a moveable piston positioned within the cylinder, the piston including a first side and second side opposite the first side, the second side being exposed to the pressurized gas in the cylinder and storage chamber cylinder; a driver blade attached to the piston and movable therewith between a top-dead-center position and a bottom-dead-center position, the driver blade defining a driving axis; a lifter operable to move the driver blade from the bottom-dead-center position toward the top-dead-center position, the lifter is configured to engage the driver blade when moving the driver blade from the bottom-dead-center position toward the top-dead-center position; and a sliding seal disposed between the piston and the cylinder; wherein during movement of the piston and driver blade from the top-dead-center position to the bottom-dead-center position, a pressure on the second side of the piston is greater than a pressure on the first side of the piston and the sliding seal is configured to prevent pressurized gas from passing through an annular space between the piston and the cylinder, and

wherein as the piston approaches the bottom-dead-center position and the pressure on the first side of the piston exceeds the pressure on the second side of the piston, the sliding seal is configured to permit a bypass flow of pressurized gas to pass in a direction from the first side of the piston through an annular space between the piston and the cylinder into a space within the cylinder adjacent the second side of the piston, thereby increasing a pressure of the pressurized gas within the cylinder and storage chamber cylinder, wherein the sliding seal includes a ring and an arm, the arm being integrally formed with and extending from the ring, wherein the ring extends inward into a groove formed in the piston, and wherein the arm is movable relative to an outer circumferential surface of the piston.

10. The gas spring-powered fastener driver of claim 9, wherein the arm is a first arm, wherein the sliding seal includes a second arm integrally formed with and extending from the ring, and wherein the first arm and the second arm are angled away from each other.

11. The gas spring-powered fastener driver of claim 10, wherein the bypass flow of pressurized gas causes the first arm to move toward the second arm, thereby creating the annular space between the piston and the cylinder.

12. The gas spring-powered fastener driver of claim 9, further comprising a bumper positioned beneath the piston in a vertical direction to absorb impact energy from the piston, and wherein as the piston approaches the bottom-dead-center position, the piston compresses the bumper causing the pressure on the first side of the piston to exceed the pressure on the second side of the piston.

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